

CUSTOMER ENGAGEMENT & PRODUCT UTILIZATION ANALYTICS FOR RETENTION STRATEGY

Executive Summary for Retention Stakeholders

Data Analytics | Customer Retention | Streamlit | Machine Learning

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Decision Summary

The bank retains 79.63% of customers in the historical dataset, but churn rises sharply when relationships become inactive or product configuration is weak. Inactive customers churn at 26.85%; active customers churn at 14.27%. Germany has the highest market churn at 32.44%.

Most Important Findings

Finding	Evidence	Recommended action
Inactive customer risk	26.85% churn	Launch reactivation journeys
Premium disengagement	30.47% churn among inactive premium customers	Relationship-manager priority queue
One-product weakness	27.71% churn	Needs-based second-product offers
Two-product strength	7.58% churn	Use as relationship-depth benchmark
Germany priority	32.44% churn	Localized root-cause pilot
Credit-card signal	20.18% vs 20.81%	Do not rely on ownership alone

90-Day Action Plan

Period	Action
0-30 days	Validate premium risk list, assign owners, review Germany and 3+/4-product customer cases.
31-60 days	Run controlled reactivation and one-to-two-product pilots with holdout groups.
61-90 days	Measure retained balances, response, complaints, fairness outcomes and incremental retention.

Governance

- Use predictions to prioritize human review, not to automate adverse customer decisions.
- Gender is excluded from operational prediction and retained for fairness monitoring.
- Validate extreme product-count findings because the 3- and 4-product groups are small.
- Track intervention lift against untreated control groups before scaling.

Delivered System

The Streamlit dashboard provides live filters, executive KPIs, engagement and product analysis, premium-risk mapping, a downloadable priority queue, model performance, feature importance, a customer risk simulator and searchable customer records.